

A Green Temperature-Responsive Hydrogel Derived from Rosin-Stabilized N-Vinylcaprolactam and Agrochemicals for Synergistic Pesticide-Fertilizer Release

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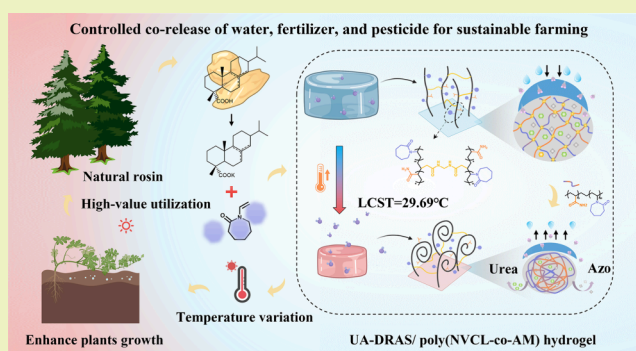
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ABSTRACT: The practical use of agricultural hydrogels remains limited by poor mechanical strength, incompatibility with hydrophobic agrochemicals, and weak environmental responsiveness, although they have advanced in retaining water and delivering nutrients. In this study, a thermoresponsive hydrogel was synthesized, with urea incorporated as a nutrient source, by copolymerizing the hydrophobic monomer N-vinylcaprolactam (NVCL) with the fungicide azoxystrobin (Azo), using amphiphilic disproportionated rosin acid soap (DRAS) as both an emulsifier and a structural modifier. This dual role of DRAS enabled it to not only disperse the hydrophobic components but also regulate the hydrogel's network architecture, thereby enhancing its mechanical strength, swelling behavior, and controlled release kinetics. Below the lower critical solution temperature (LCST), the hydrogel demonstrated effective soil moisture retention and reduced water evaporation. Upon heating above the LCST, it facilitated the controlled release of urea and Azo compounds. This temperature-responsive behavior allows for the on-demand coordination of water, fertilizer, and pesticide delivery in alignment with crop growth stages and disease prevention needs. Biosafety evaluation revealed that the unloaded hydrogel exhibited low toxicity toward earthworms. In antifungal tests, the drug-loaded hydrogel showed strong inhibitory activity against *Fusarium oxysporum* f. sp. *niveum* (FON). Pot experiments further confirmed that the drug-loaded hydrogel effectively suppressed FON infection, increased watermelon biomass, and improved fruit quality. These results highlight the potential of this multifunctional hydrogel as an environmentally sustainable and stimulus-responsive carrier for smart agrochemical delivery in modern agriculture.

KEYWORDS: *N-vinylcaprolactam, amphiphilic rosin, thermoresponsive hydrogel, hydrophobic pesticide, integrated agrochemical delivery, sustainable agriculture*



INTRODUCTION

Within the complexity of agricultural ecosystems, crops often face concurrent stresses, including drought, pest infestations, and nutrient deficiencies.^{1–3} To address this issue, researchers have developed various controlled-release carriers for agricultural chemicals, such as nanoparticles, microcapsules, emulsions, and porous microspheres.^{4–9} However, these systems primarily target the aerial parts of plants and offer limited efficacy in modulating the rhizosphere microenvironment.¹⁰ In comparison, hydrogels exhibit excellent water retention capacity and stimuli-responsive release behavior.^{11,12} They can deliver water and agrochemicals directly to the root zone, making them a promising platform for soil improvement and smart fertilization.^{13,14} Nevertheless, in complex agricultural ecosystems involving multiple interacting stresses, conventional hydrogels remain limited in function. They usually offer only a single-agent release of either fertilizer^{15–18} or pesticide,^{19,20} which falls short of the integrated soil management needed for sustainable crop production. These challenges

underscore the need for multifunctional, environmentally responsive hydrogels for smart and adaptive control of agrochemical release.

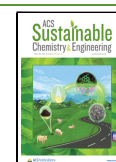
Among various environmental cues, temperature is highly synchronized with crop growth rhythms and disease occurrence, making it an ideal regulatory factor for achieving precise delivery.²¹ Temperature-responsive hydrogels exploit the reversible phase transition of their polymer networks to release agrochemicals at elevated temperatures.^{22–24} This behavior ensures that the release aligns with periods of nutrient demand and peak disease incidence in crops.

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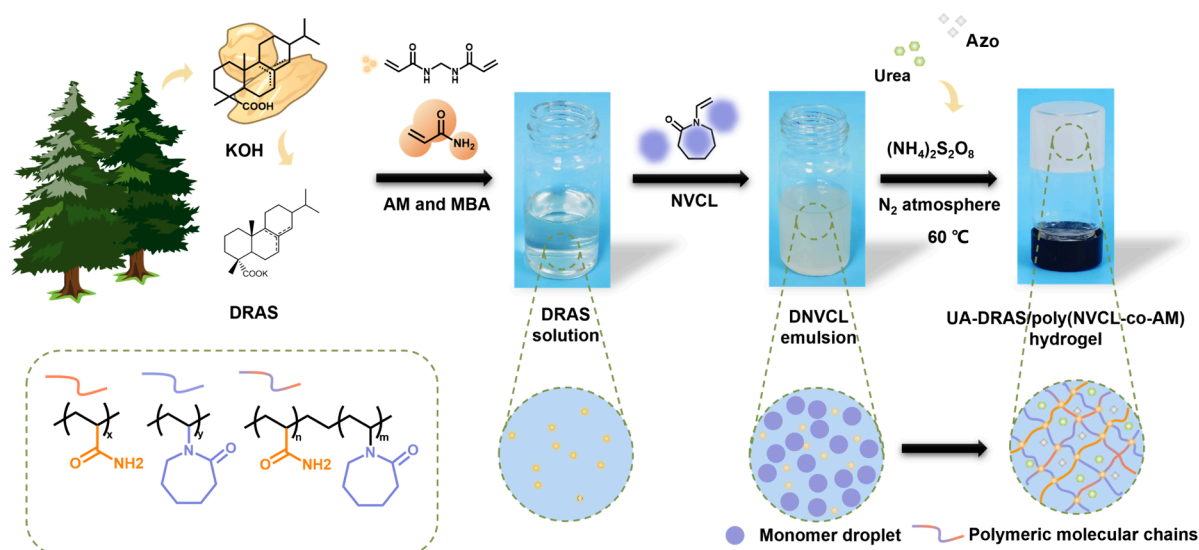


Figure 1. Synthesis process of the UA-DRAS/poly(NVCL-co-AM) hydrogel. This hydrogel integrates temperature-responsive poly(NVCL-co-AM), hydrophobic DRAS domains, and urea loading, enabling synergistic control of water retention, nutrient release, and pesticide delivery for sustainable agricultural applications.

Although *N*-isopropylacrylamide (NIPAM) is a typical thermoresponsive polymer, its use in agriculture is limited due to the potential generation of toxic byproducts during amide bond hydrolysis.^{4,25,26} In contrast, NVCL possesses a stable six-membered lactam ring and exhibits excellent biocompatibility, making it more suitable for agricultural applications.^{27,28} NVCL-based hydrogels have been shown to perform well in soil water retention and controlled fertilizer release.^{29–32} However, their strong hydrophobicity often leads to poor dispersibility and insufficient mechanical strength, hindering large-scale field applications.

DRAS is an alkaline salt of disproportionated rosin (DPR), mainly composed of dehydroabietic acid moieties.³³ Its amphiphilic structure makes it a widely used emulsifier in fields such as food, agrochemicals, and the rubber industry.^{34–36} Importantly, DRAS possesses a rigid tricyclic phenanthrene skeleton. In polymer hydrogels, such rigid molecular structures are well-known for their ability to significantly enhance mechanical strength and structural stability.^{37,38} For instance, the incorporation of dehydroabietic acid into hemicellulose-based hydrogels has been reported to markedly improve their mechanical properties through synergistic hydrophobic interactions, hydrogen bonding, and π - π stacking.³⁹ In addition, DRAS is naturally derived, environmentally compatible, and low-cost, making it highly suitable for large-scale adoption in sustainable agriculture.³⁷ Therefore, incorporating DRAS into thermoresponsive hydrogel systems allows us to leverage its unique combination of properties to construct high-performance, sustainable carriers for synchronized water and agrochemical delivery.

In this work, we dispersed the hydrophobic monomer NVCL and pesticide Azo in water using the natural emulsifier DRAS, then copolymerized the mixture with acrylamide (AM) and urea via free-radical polymerization to form a thermoresponsive hydrogel (UA-DRAS/poly(NVCL-co-AM)). The effects of DRAS content on the microstructure, mechanical strength, swelling behavior, and water retention capacity were systematically studied. The thermoresponsive release performance of urea and Azo was evaluated. In addition, the biosafety of the unloaded hydrogel was evaluated using earthworm

assays. The antifungal activity of Azo-loaded hydrogel against FON was tested in vitro. Pot experiments with watermelon were conducted to examine disease prevention efficacy and plant growth promotion. Overall, this study proposes a promising platform for the development of multifunctional controlled-release carriers for water, fertilizers, and pesticides.

MATERIALS AND METHODS

Materials

Disproportionated rosin (dehydroabietic acid >52%) was obtained from Guangxi Jinxiu Pine Wood Products Co. Potassium hydroxide (KOH, 85%), ammonium persulfate (APS, 98%), *N*-vinylcaprolactam (NVCL, 99%), and acrylamide (AM, 99%) were sourced from Aladdin Chemical Co. the cross-linker *N*, *N*'-methylenebis(acrylamide) (MBA, 99%) were provided by Shanghai Mai Lin Biochemical Co. Urea (M_w 60.06 g/mol, white needle crystals) and azoxystrobin (Azo, 98.2%) were purchased from Nanjing Chemical Reagent Co. and Taizhou Baili Chemical Co., respectively. The test strain of watermelon wilt, *Fusarium oxysporum* f. sp. *niveum*, was provided by the School of Plant Protection, Northwest Agriculture and Forestry University.

Preparation of DRAS

60 g of 10% KOH aqueous solution was added to a three-necked round-bottom flask. The system was heated to 85 °C. Then, 30.2 g of DPR was gradually added under stirring at 200 rpm, and the reaction was continued for 3 h to give DRAS. The product was subsequently diluted to solid concentrations of 4, 4.5, 5, and 5.5% for further application.

Synthesis of DRAS/Poly(NVCL-co-AM) Hydrogel

The synthesis procedure for the DRAS/poly(NVCL-co-AM) nanocomposite hydrogel is schematically illustrated in Figure 1. AM and MBA were first dissolved in aqueous DRAS solutions. The mixture was then charged with NVCL at 40 °C under stirring, producing stable monomer emulsions. After the ice-bath addition of urea, Azo, and APS, the mixture was subjected to polymerization under a nitrogen atmosphere at 60 °C for 3 h, yielding the final hydrogel. The detailed formulations were provided in Table S1. The effects of the NVCL-to-AM ratio, cross-linker and initiator concentrations, and DRAS content were also investigated, with detailed results and discussion provided in the Supporting Information (Figure S1).

■ CHARACTERIZATION

Fourier Transform Infrared Spectrometry

FTIR spectra of DRAS, NVCL, AM, and hydrogels with different DRAS loadings were acquired in ATR mode on a Nicolet ISSO spectrometer (Thermo Fisher Scientific) over 4000–600 cm^{-1} at 1 cm^{-1} resolution.

Scanning Electron Microscopy (SEM)

Surface morphology of freeze-dried hydrogel samples was examined using a JEOL JSM-7600F field-emission SEM at 5 kV after mounting on carbon tape (sputter-coating applied when required).

Mechanical Properties

Compressive and cyclic compression tests (up to 80% strain) were performed on a UTM6502 universal testing machine (Suns, China) to evaluate mechanical strength and elastic recovery.

Differential Scanning Calorimetry (DSC)

The hydrogel samples were tested using a TA Q2000 and heated from 10 to 70 $^{\circ}\text{C}$ at 2 $^{\circ}\text{C}/\text{min}$, and the LCST was determined from the glass transition temperature (T_g) of the endothermic transition.

Swelling Properties and Kinetics

Swelling and thermally induced deswelling of the hydrogels were tested as previously described.³¹ Samples of known dry mass were immersed in deionized water at 20 $^{\circ}\text{C}$, and mass measurements were taken at predetermined intervals until swelling equilibrium was reached. The swelling ratio (SR) can be calculated in triplicate using eq 1 below.

$$\text{SR} = \frac{W_t - W_0}{W_0} \times 100\% \quad (1)$$

Where W_0 corresponds to the initial dry weight, and W_t refers to the weight of the swollen hydrogel at time t (h).

Shrinking Kinetics

Fully swollen hydrogels were transferred to deionized water at 50 $^{\circ}\text{C}$. At predetermined time intervals, each sample was removed and gently blotted dry with filter paper. Each sample was quickly weighed, and its water retention (WR) was computed from eq 2.

$$\text{WR} = \frac{(W_h - W_0)}{W_s} \times 100\% \quad (2)$$

Where W_h represents the mass of the hydrogel at time t at 50 $^{\circ}\text{C}$, W_0 is the dry weight of the hydrogel, and W_s denotes the equilibrium swollen weight measured at 20 $^{\circ}\text{C}$.

Thermoresponsive Behavior and Cyclic Performance

The method for determining the equilibrium swelling ratio (ESR) was adopted from the literature.³¹ Briefly, the swelling behavior of the hydrogels was characterized gravimetrically after equilibrating them in deionized water across a temperature range of 10 to 50 $^{\circ}\text{C}$ for 24 h. Moreover, cyclic equilibrium swelling tests were performed to evaluate the switching behavior of the hydrogels between 20 and 50 $^{\circ}\text{C}$ for four complete cycles. ESR was calculated using eq 3 below.

$$\text{ESR} = \frac{(W_e - W_0)}{W_0} \times 100\% \quad (3)$$

Where W_e represents the mass of the hydrogel at equilibrium swelling, and W_0 denotes the dry mass.

Swelling Kinetics of Hydrogel

The swelling kinetics of the hydrogels at 20 $^{\circ}\text{C}$ were fitted using the Korsmeyer-Peppas model and the pseudo-second-order kinetic model.^{29,40} These models are expressed as eqs 4 and 5 below.

$$F = \frac{Q_t}{Q_e} = k_1 t^n \quad (4)$$

$$\frac{t}{Q_t} = \frac{1}{k_2 Q_e^2} + \frac{1}{Q_e} t \quad (5)$$

In these models, Q_t represents the water content absorbed at time t , Q_e is the water content at equilibrium, k_1 is the rate constant, n is the diffusional exponent, and k_2 is the pseudo-second-order swelling constant.

Soil Water-Retaining Capacity of Hydrogel

The soil water-retaining ratio (WR) of soil amended with hydrogel was evaluated according to a literature method.³⁰ In brief, a mixture of 99 g of soil and 1 g dry hydrogel was water-saturated, and its weight loss was tracked alongside a soil-only control over 30 days at ambient conditions. Equation 6 was applied to calculate the WR ratio from the recorded weights.

$$\text{WR} = \frac{(W_t - W)}{(W_0 - W)} \times 100\% \quad (6)$$

Where W_t represents the mass of the hydrogel-soil mixture measured at time t under ambient conditions, W_0 is the initial mass of the dry hydrogel-soil mixture, and W_s denotes the weight of the bare soil.

Controlled Release Behavior in Water and Soil

Before release experiments, calibration curves for urea and Azo were determined by UV spectrophotometry and liquid chromatography (Figure S2). Subsequently, 5 g of hydrogel loaded with 5% Azo and 50% urea was incubated in 100 mL of deionized water at 20 and 40 $^{\circ}\text{C}$. The cumulative release over time was monitored using eq 7.

$$\text{Cumulative release} = \frac{m_t}{m_1} \times 100\% \quad (7)$$

Where m_t refers to the mass of urea or Azo released at time t , and m_1 represents the total initial loading of the respective agent in the hydrogel.

Soil release studies were conducted by applying 1 g of UA-D_{5.5}/NA hydrogels to 100 g of dry soil. The tests investigated the effects of moisture (15 and 35%) at 30 $^{\circ}\text{C}$ and temperature (20 and 40 $^{\circ}\text{C}$) on Azo and urea release over 30 days. Soil samples were periodically collected, and the residual agents in the hydrogels were extracted with a methanol/water (75:25, v/v) solution for quantification via standard curves. Cumulative release was determined as the difference between the initial load and the residual amount. Total soil nitrogen was determined using the Kjeldahl method.

Antifungal Studies of Hydrogel

The efficacy of UA-D_{5.5}/NA against FON was evaluated using the mycelial growth rate method.⁴¹ The hydrogel leachate, collected by soaking in deionized water at 40 $^{\circ}\text{C}$ and quantified via standard curves, was blended into sterilized PDA medium. A mycelial plug was then inoculated onto the amended

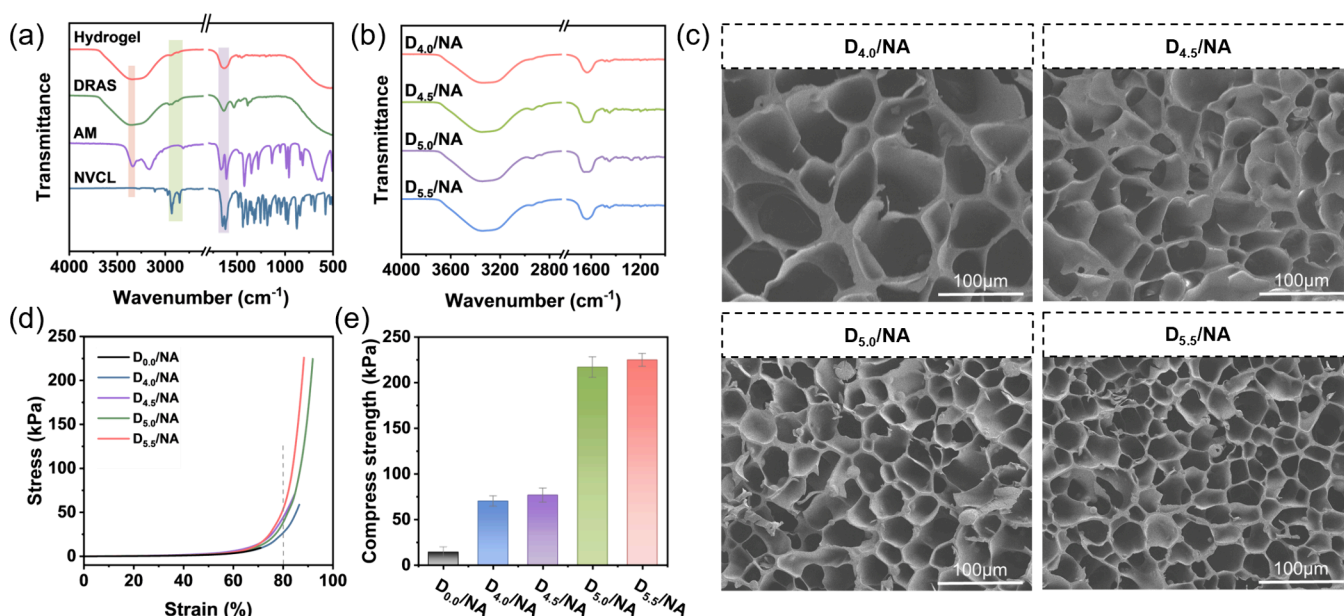


Figure 2. (a) FTIR analysis of raw materials (NVCL, AM, DRAS) and the resulting hydrogel. (b) FTIR analysis of hydrogels with varying DRAS concentrations. (c) SEM images of hydrogels with varying DRAS concentrations. (d) Compressive stress–strain profiles and (e) derived strength across various DRAS contents.

medium. After incubation, once fungal colonies reached 7–8 cm in diameter, the inhibition zone was measured by the cross method, and the inhibition rate (IR) was derived from eq 8.

$$IR = \frac{(\varnothing_0 - \varnothing_t)}{\varnothing_0} \times 100\% \quad (8)$$

where $\varnothing_0$ and $\varnothing_t$ are the colony diameters (minus the 5 mm diameter of the mycelial plug) of the CK and treatment groups, respectively.

Pot Experiment

Uniform seedlings were transplanted into 15 cm-diameter PVC pots filled with nitrogen-free soil. After 15 days of acclimation, five treatments were established: (1) control (nitrogen-free soil), (2) D_{5.5}/NA hydrogel, (3) 1 g urea, (4) commercial slow-release nitrogen fertilizer (0.46 g N), and (5) UA-D_{5.5}/NA hydrogel (0.46 g N). All treatments were watered with tap water to 70% of water-holding capacity and irrigated 2–3 times per week to maintain this level. Plant growth was assessed by measuring vine length and fruit diameter at 60, 75, and 90 days after treatment. At harvest (90 days), the roots and vines were thoroughly rinsed with distilled water, oven-dried, and their biomass was determined.

Disease Resistance

Plants were wounded at the roots and inoculated at the root-stem junction with FON (10^6 spores/mL, 10 mL per plant). Simultaneously, UA-D_{5.5}/NA hydrogel (30 mg Azo per pot) was embedded in the rhizosphere. Disease development was visually assessed and photographed after inoculation.

Biosafety Evaluation

The biosafety of D_{5.5}/NA was further assessed using an earthworm toxicity test. Ten earthworms were introduced into pots containing 40 g of soil amended with different concentrations of D_{5.5}/NA. Each system was moistened with 50 mL of water and maintained at adequate moisture by irrigation 2–3 times per week. After a 15-day incubation

period, earthworm viability was calculated using the equation below (eq 9).⁴⁴

$$Viability = \frac{N_t}{N_0} \times 100\% \quad (9)$$

Where N_t and N_0 represent the number of surviving and initial earthworms, respectively.

Moreover, a two-chamber avoidance test was conducted to evaluate the behavioral response of earthworms to hydrogel-amended soil. A divided container with a removable central partition was used, and small openings were made at the bottom to allow earthworm movement between chambers. Each chamber was filled with 20 g of soil. One chamber contained soil amended with hydrogel at 0, 2.20, 4.49, or 8.98 mg/g, while the other contained untreated soil. Ten healthy earthworms with similar body weights were initially placed in the treated chamber. After 10 days of incubation under controlled conditions, earthworm survival, body weight changes, and distribution between chambers were recorded. The avoidance rate (AR) was calculated using the equation below (eq 10).

$$AR = \frac{N_c - N_t}{N} \times 100\% \quad (10)$$

where N_c represents the number of earthworms in the control chamber, N_t represents the number of earthworms in the treated chamber, and N is the total number of earthworms.

RESULTS AND DISCUSSION

The Dispersion Stability of NVCL in Aqueous Solution

The hydrophobic nature of NVCL hinders the direct formation of homogeneous hydrogels in aqueous solution. To address this, DRAS functioned as a biobased emulsifier, enabling the creation of a stable emulsion. As shown in Figure S3, the incorporation of DRAS effectively dispersed NVCL, producing a uniform emulsion, while pronounced phase

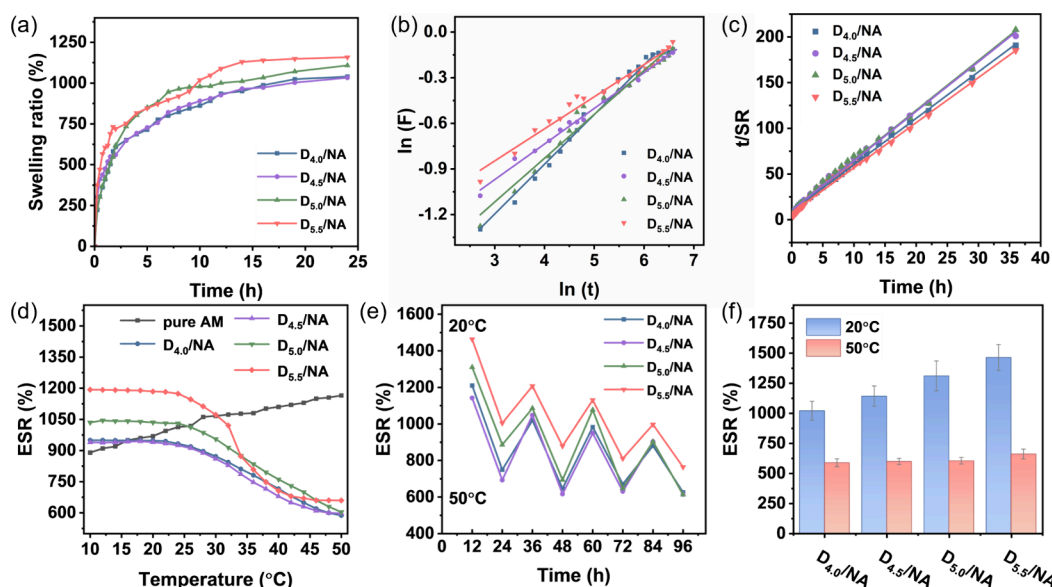


Figure 3. (a) Swelling ratios of different hydrogels in distilled water at 20 °C. (b) Korsmeyer-Peppas and (c) Schott second-order kinetic modeling of the swelling process. (d) Equilibrium swelling ratios of hydrogels at different temperatures. (e) Cyclic swelling–deswelling behavior of hydrogels with varying DRAS content. (f) Equilibrium swelling ratios at 20 and 50 °C.

separation occurred in the absence of DRAS. The droplet size distribution (Figure S4) further confirmed this effect, showing a gradual reduction in emulsion droplet size with increasing DRAS content. This behavior arises because increased DRAS content enhances interfacial adsorption and coverage.⁴³ As a result, a more compact interfacial film is formed, which resists droplet coalescence and leads to smaller, more stable droplets. These results demonstrate that DRAS not only stabilizes the emulsion effectively but also enables tunable control over droplet size.

Preparation and Characterization of Hydrogels

The FTIR spectrum of NVCL (Figure 2a) exhibited a strong C=C stretching vibration at 1648 cm^{-1} , along with characteristic vinyl group signals at 3107 cm^{-1} (=C–H) and 991 cm^{-1} (=CH₂). The complete disappearance of these characteristic double-bond peaks confirms successful covalent cross-linking.⁴⁴ A distinct doublet was observed for AM, corresponding to the N–H stretches of the primary amide group at 3338 cm^{-1} (asymmetric) and 3173 cm^{-1} (symmetric).²⁹ In the hydrogel spectrum, these distinct peaks merge into a single broad band centered at 3340 cm^{-1} . This broadening and shift indicate the integration of the amide groups into a hydrogen-bonded network with surrounding water molecules and the carboxylate group of DRAS.³⁷ An increase in DRAS content led to a slight enhancement in both the intensity and bandwidth of the peak at 3340 cm^{-1} (Figure 2b), suggesting a strengthening of the hydrogen-bonding interactions within the hydrogel network.⁴⁵ Collectively, these spectral shifts demonstrate successful monomer incorporation and network formation.

SEM images (Figure 2c) revealed the internal morphology of the lyophilized hydrogels with different DRAS contents. All samples exhibited a well-developed porous structure, which is crucial for efficient agrochemical delivery. As the DRAS content increased, the pore size gradually decreased, consistent with the observed reduction in prepolymer emulsion droplet size. Smaller droplets bring the dispersed phases into closer contact, facilitating the formation of a more compact and

continuous network framework, as illustrated in Figure S5 and reported previously.³² These results indicate that DRAS acts as a pore-structure regulator, promoting a finer, more uniform microstructure.

Mechanical Properties of Hydrogels

Compression and cyclic compression tests were performed on samples with different DRAS contents. Under compression testing, the DRAS-free hydrogel (D_{0.0}/NA) exhibited a maximum compressive strength of 14.0 kPa at 72.3% strain (Figure 2d). And a clear trend of increasing compressive strength was observed with rising DRAS content. The hydrogel with 5.5% DRAS (D_{5.5}/NA) achieved the highest compressive strength of 225 kPa, which is 16.1 times higher than that of D_{0.0}/NA (Figure 2e). This mechanical enhancement primarily originates from the unique molecular structure of DRAS.³⁹ Its rigid hydrophenanthrene rings establish reversible physical cross-links with NVCL chains through hydrophobic interactions, synergizing with the chemical cross-links formed by MBA to reinforce the network. Moreover, DRAS promotes a more uniform and denser porous structure, which collectively improves the mechanical performance. Moreover, as the DRAS content increased from 4.0 to 5.5%, the compressive modulus rose from 71.9 to 433.6 kPa, accompanied by an increase in dissipated energy from 0.49 to 1.63 kJ/m^3 (first cycle) and from 0.54 to 1.96 kJ/m^3 (10th cycle). To further elucidate the mechanical reinforcement mechanism, rheological analyses (Figure S6) were performed. These results indicate that DRAS-induced reversible physical cross-links, together with a reduced effective chemical cross-linking density, promote network reconstruction under cyclic loading, thereby enhancing polymer ordering and energy dissipation efficiency.⁴⁶

Elucidation of Swelling Dynamics and Mechanisms

As shown in Figure 3a, the ESR increased linearly with DRAS content, rising from 1020% (D_{4.0}/NA) to 1463% (D_{5.5}/NA). This trend can be attributed to the thermoresponsive behavior of NVCL-based hydrogels, which undergo a volume phase transition at the LCST driven by polymer chain rearrangements. Below the LCST, hydrogen bonding promotes chain

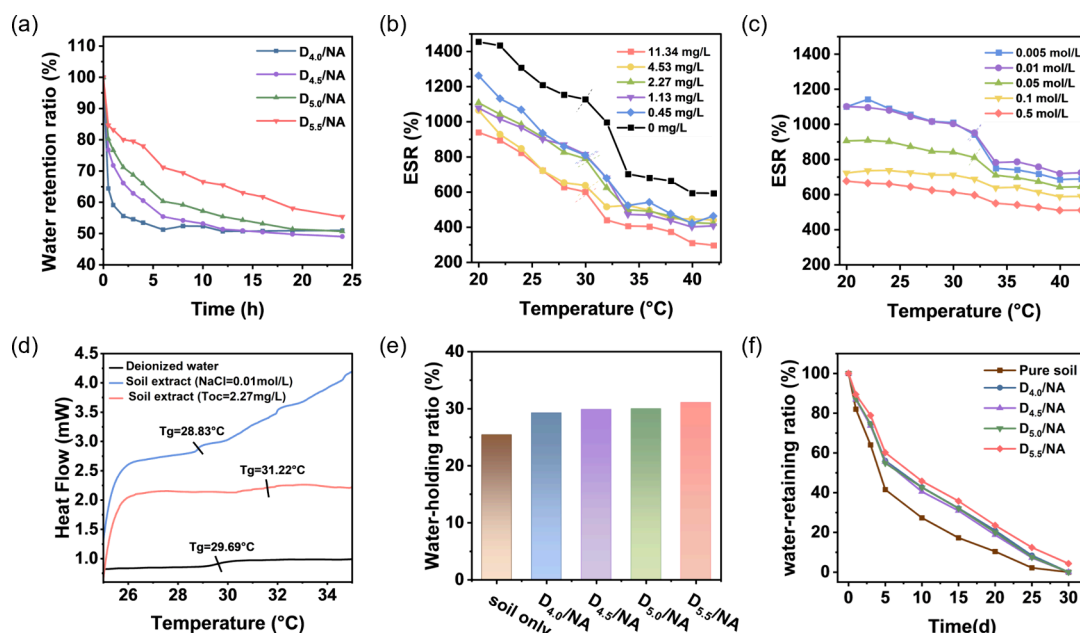


Figure 4. (a) Water retention ratios of various hydrogels above 50 °C. (b) Temperature-dependent ESR of the hydrogel in soil extract solutions with different total organic carbon (TOC) content and (c) ionic strength (NaCl concentration). (d) DSC thermogram of D_{5,5}/NA in deionized water and soil extract solutions. (e) Soil water-holding capacity and (f) evaporation profiles of soils amended with 1 wt % hydrogel.

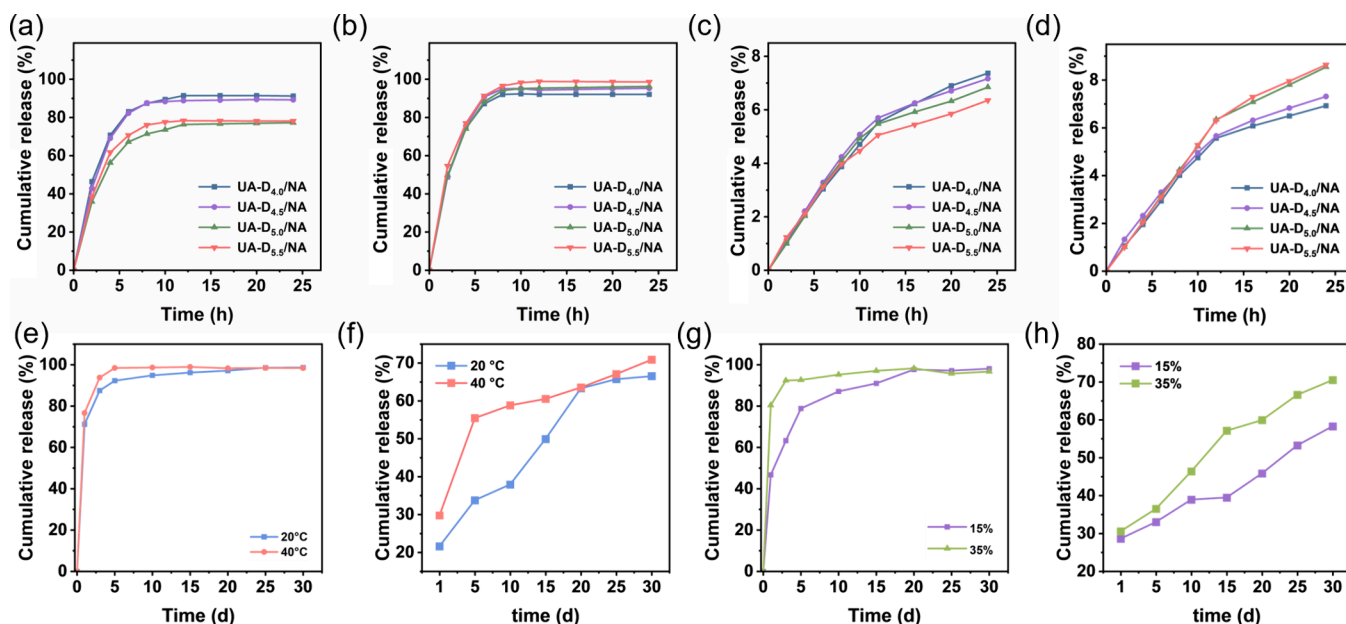


Figure 5. (a) Cumulative urea release at 20 °C and (b) 40 °C in water. (c) Cumulative Azo release at 20 °C and (d) 40 °C in water. (e) Cumulative urea and (f) Azo release in soil at different temperatures. (g) Cumulative urea and (h) Azo release in soil at different moisture content.

expansion and water uptake, whereas above the LCST, enhanced hydrophobic interactions induce chain collapse and volume contraction.⁴⁷

The swelling kinetics were further analyzed using two models.⁴⁸ Korsmeyer-Peppas plots ($\ln F$ vs $\ln t$) showed strong linearity for all hydrogels ($R^2 > 0.97$), with diffusional exponent (n) values below 0.5 (Figure 3b, Table S2). This indicates that the initial water uptake is dominated by Fickian diffusion, a process driven by a concentration gradient.⁴⁹ Schott second-order kinetic plots (t/Q_t vs t) also displayed strong linearity ($R^2 > 0.99$), as shown in Figure 3c. Swelling kinetic parameters, including R ,² k_2 , and the theoretical Q_e are

summarized in Table S2. The close match between theoretical Q_e and measured ESR values indicates that the slow swelling stage follows a pseudo-second-order kinetic model, where water transport is governed by polymer chain relaxation.²⁹

Temperature-Responsive Swelling Behavior

The hydrogels exhibited distinct thermoresponsive swelling in water. As shown in Figure 3d, all DRAS-containing hydrogels showed a reversible reduction in swelling with increasing temperature, in contrast to the nonresponsive AM hydrogel. Cyclic swelling–deswelling tests between 20 and 50 °C were used to assess hydrogel reusability. As shown in Figure 3e, swelling capacity gradually decreased over successive cycles.

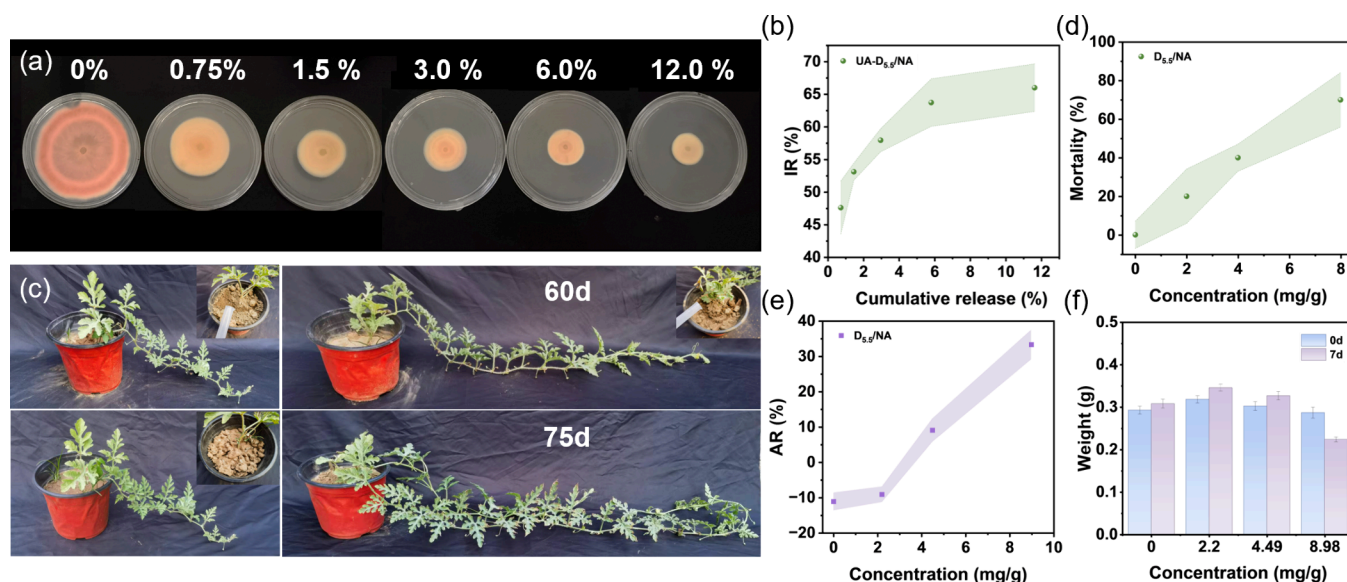


Figure 6. (a) Inhibition zone images from the antifungal assay. (b) Antifungal rate versus cumulative Azo release. (c) Representative images of watermelon plants' plant resistance. (d) Earthworm mortality, (f) weight, and (e) avoidance response in artificial soil amended with $D_{5.5}/NA$ hydrogel at different concentrations.

Hydrogels with higher DRAS content retained higher ESR and consistent performance over four cycles (Figure 3f). This improved durability arises from the rigid hydropheanthrene rings and the uniform porous network at higher DRAS content, which reduces structural damage during repeated transitions.

Water-Holding and Retention Behavior in Soil

The hydrogels displayed distinct water-retention properties at 50 °C, as revealed in Figure 4a. All hydrogels exhibited similar deswelling behavior above the LCST, with $D_{5.5}/NA$ retaining the highest water content (55.4%) at the end of the test. To further elucidate the temperature-responsive transition underlying this deswelling behavior, the LCST of the hydrogel was examined by combining temperature-dependent ESR measurements and DSC.⁵⁰ As shown in Figure 4b,c, a sharp decrease in ESR was observed with increasing temperature near the LCST, consistent with the endothermic transition detected by DSC (Figure 4d). Compared with deionized water (29.69 °C), the LCST slightly increased in soil extracts containing moderate levels of dissolved organic matter (TOC = 2.27 mg/L, 31.32 °C), whereas a marginal decrease was observed in the presence of NaCl (28.83 °C), consistent with the literature reports.⁵¹ In contrast, variations in pH and ionic species showed negligible influence on the LCST (Figure S7).

Moreover, Figure 4e shows the maximum water-holding capacity of soils containing 1 wt % freeze-dried hydrogels. The $D_{5.5}/NA$ hydrogel reached the highest water-holding ratio (31.1%), 5.7 percentage points above untreated soil (25.4%), and retained 4.3% water after 30 days (Figure 4f). These findings suggest that $D_{5.5}/NA$ exhibits outstanding water-retention performance and long-term stability, making it suitable for use in agricultural water-management systems.

Fertilizer and Pesticides Controlled-Release Properties of Hydrogel

The controlled-release performance was evaluated using hydrogels loaded with 50 wt % urea and 5 wt % Azo in aqueous and soil environments. The maximum loading capacities of urea and Azo, as well as their physical incorporation into the polymer network, confirmed by FTIR

analysis, are presented in Figure S8. The cumulative urea release profile, shown in Figure 5a,b, exhibited a two-stage profile, consisting of a rapid burst within 6 h and a slower sustained phase. This pattern arises from the rapid release of surface-bound urea and slower diffusion of urea inside the porous network.^{42,52} Moreover, urea release decreased from 91.2 to 78.2% content at 20 °C, while it rose from 89.1 to 99.4% at 40 °C with higher DRAS. A similar trend was observed for Azo, decreasing from 7.4 to 6.4% at 20 °C and increasing from 7.0 to 8.6% at 40 °C (Figure 5c,d). This reversal in release behavior is governed by the DRAS content. Below the LCST, higher DRAS content forms a denser network with stronger hydrogen bonding that limits diffusion. Above the LCST, the hydrophilic-to-hydrophobic transition collapses the network, and the hydrophobic DRAS domains further accelerate release (Figure S9).

As shown in Figure 5e,f, elevated temperature (40 °C) promoted higher cumulative release of both urea and Azo, confirming the thermoresponsive behavior observed in soil. Urea was rapidly released within 3 days, which corresponded to a sharp increase in soil nitrogen content. Moreover, reversible swelling–deswelling and temperature-synchronized release were observed under simulated diurnal temperature cycling, demonstrating the good repeatability of the thermoresponsive behavior (Figure S10). Azo exhibited a slower, more sustained release profile, with a cumulative release of 70.9% at 40 °C after 30 days, due to its confinement within hydrophobic domains in the hydrogel network (Figure S11). The effect of soil moisture on the release behavior was further investigated. Higher soil moisture levels promoted faster and greater cumulative release of both urea and Azo.

As shown in Figure 5g, urea was completely released after 30 days, with the soil nitrogen content increasing to 0.43 and 0.18 g/kg at 15 and 35% soil moisture, respectively (Figure S10). Meanwhile, the cumulative release of Azo reached 58.2 and 70.5% under the same conditions (Figure 5h). This enhancement can be attributed to improved water exchange and molecular diffusion within the hydrogel matrix at higher

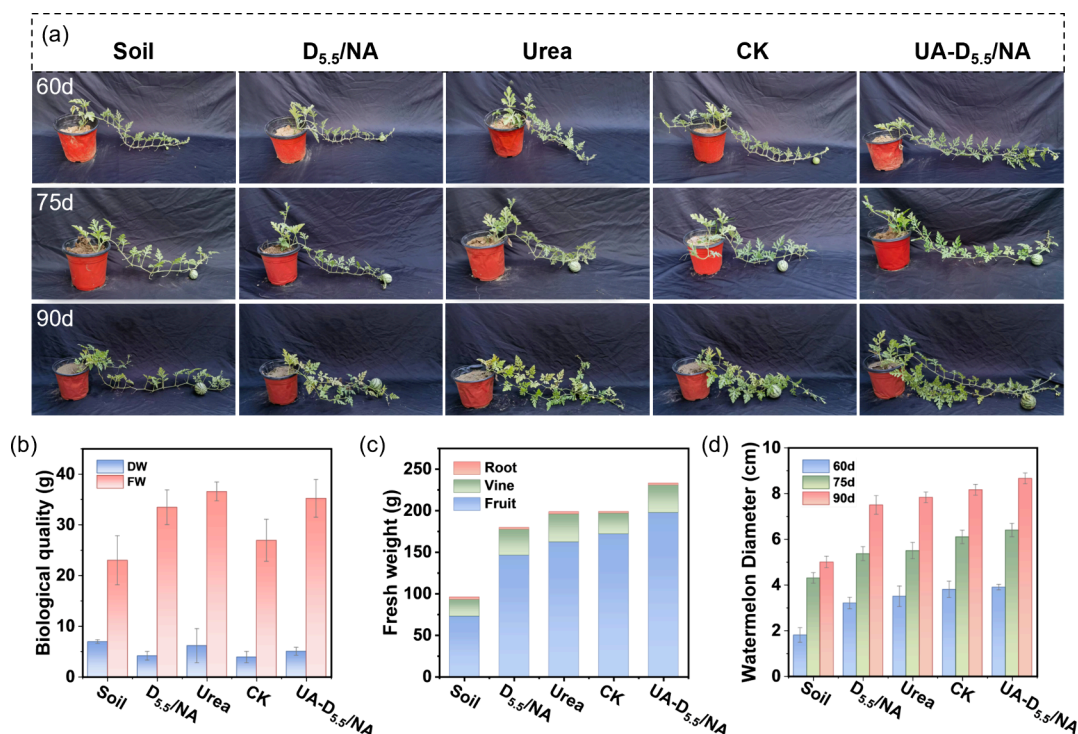


Figure 7. (a) Images of watermelon plants under different treatments. (b) Dry and fresh biomass of watermelon plants at harvest. (c) Biomass distribution of roots, vines, and fruits. (d) Fruit diameter at 60, 75, and 90 days.

moisture levels, which facilitates network relaxation and accelerates the release of nutrients and pesticides.⁵³

Antifungal Efficacy and Biosafety of Hydrogels

The antifungal performance of the UA-D_{5.5}/NA hydrogel against FON was assessed using an *in vitro* agar diffusion assay (Figure 6a,b). The leachate of UA-D_{5.5}/NA produced a distinct inhibition zone, whose diameter gradually decreased with the cumulative release of Azo. The antifungal rate reached 66% at a cumulative release of 12%, confirming that the bioactivity of Azo was well preserved during encapsulation and release. *In vivo* resistance assays using watermelon plants (Figure 6c) further supported these findings. Plants pretreated with UA-D_{5.5}/NA showed no disease symptoms even after 15 days of inoculation, whereas untreated plants displayed severe leaf wilting. These results demonstrate that the UA-D_{5.5}/NA hydrogel effectively protects crops from pathogen infection.

The environmental safety of the carrier was also examined through an artificial soil assay using earthworms as bioindicators.⁴² Soils amended with D_{5.5}/NA showed limited effects on earthworm viability at low concentrations. Even at a high application rate of 2.2 mg/g soil, the earthworm mortality rate was 30% (Figure 6d), indicating acceptable biosafety under practical use conditions.⁵⁴ Moreover, a two-chamber avoidance test revealed no significant avoidance behavior (AR < 10%) or weight loss at agronomically relevant amendment levels (≤4.49 mg/g soil), whereas increased avoidance behavior and body weight reduction were observed at 8.98 mg/g soil (Figure 6e,f). Figure S12 further shows that hydrogel amendment had no detectable negative effects on root development. Consistently, Heatmap analysis together with the relative abundance of the top 20 bacterial genera indicates that both D_{5.5}/NA and UA-D_{5.5}/NA treatments caused only moderate, genus-specific shifts relative to CK, with slight enrichment of several taxa associated with organic matter

turnover and nitrogen cycling. In addition, long-term soil incubation showed that the hydrogel undergoes gradual degradation while retaining structural integrity and reproducible thermoresponsive swelling behavior, indicating sufficient stability under soil-relevant conditions (Figure S13). Overall, the UA-D_{5.5}/NA hydrogel enables controlled pesticide release for disease suppression, provides sustained nitrogen nutrition to support plant growth, and demonstrates good environmental safety.

Practical Applications of Hydrogels

Watermelon seedlings were used to evaluate the effects of UA-D_{5.5}/NA hydrogel on nitrogen supply and plant growth. After 90 days of cultivation with equal nitrogen input, plants treated with UA-D_{5.5}/NA showed significantly enhanced growth compared with the other treatments (Figure 7a). Quantitative measurements at harvest (Figure 7b) showed that both D_{5.5}/NA and UA-D_{5.5}/NA treatments resulted in higher shoot and root biomass compared than the commercial slow-release nitrogen fertilizer (CK). Analysis of biomass partitioning (Figure 7c) indicated that fruits constituted the largest fraction of total biomass. The UA-D_{5.5}/NA treatment resulted in the greatest fruit weight, followed by the commercial fertilizer group. Consistently, fruit diameter measurements throughout the growth period (Figure 7d) showed that UA-D_{5.5}/NA-treated plants maintained larger fruit size than the other treatments, indicating sustained growth during fruit development. Overall, these results indicate that the UA-D_{5.5}/NA hydrogel effectively promotes watermelon growth and fruit development through the combined effects of sustained nitrogen supply, water retention in the root zone, and disease suppression, demonstrating its potential for sustainable crop production.

In summary, we developed a temperature-responsive UA-DRAS/poly(NVCL-co-AM) as a multifunctional platform for

sustainable agriculture. The amphiphilic DRAS facilitated uniform dispersion of NVCL and Azo, which reinforced the polymer network and yielded a denser microstructure with enhanced mechanical properties, achieving a compressive strength of 225 kPa and a modulus of 433.6 kPa. The hydrogel demonstrated excellent water retention, mechanical resilience, and dual controlled release of urea and Azo. Below the LCST, hydrogen bonding enhanced water retention and restricted nutrient release, whereas above the LCST, the hydrophilic-to-hydrophobic transition accelerated the release of active agents. Biocompatibility assessments and pot experiments confirmed that the hydrogel effectively suppressed soil-borne pathogens and provided temperature-responsive nutrient release, thereby promoting plant growth and increasing yield.

■ ASSOCIATED CONTENT

Data Availability Statement

The data sets used or analyzed during the current study are available from the corresponding author on reasonable request.

■ Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acssuschemeng.5c12720>.

Additional experimental details, materials and methods, characterization data (FT-IR, CLSM, GC-MS), mechanical and rheological measurements, swelling measurements, release studies, root growth experiments, degradation experiments, and biosafety assessment (PDF)

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Notes

The authors declare no competing financial interest.

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■ REFERENCES

- (1) Savary, S.; Willocquet, L.; Pethybridge, S.; Esker, P.; McRoberts, N.; Nelson, A. The global burden of pathogens and pests on major food crops. *Nat. Ecol. Evol.* **2019**, *3* (3), 430–439.
- (2) Wu, H.; Su, X.; Huang, S.; Singh, V.; Zhou, S.; Tan, X.; Hu, X. Decreasing dynamic predictability of global agricultural drought with warming climate. *Nat. Clim. Change.* **2025**, *15* (4), 411–419.
- (3) Abdo, A.; Sun, D.; Shi, Z.; Abdel-Fattah, M.; Zhang, J.; Kuzyakov, Y. Conventional agriculture increases global warming while decreasing system sustainability. *Nat. Clim. Change.* **2025**, *15* (1), 110–117.
- (4) Xiang, N.; Lin, Y.; Qin, Z.; Su, T.; Xie, X.; Ji, H. Temperature-responsive gating chitosan-based microcapsules for controlled release of urea fertilizers. *Carbohydr. Polym.* **2025**, *348*, No. 122929.
- (5) Liu, M.; Cao, Y.; Li, Z.; Wang, E.; Ram, R.; Marelli, B. Precise and high-throughput delivery of micronutrients in plants enabled by pollen-inspired spiny and biodegradable microcapsules. *Adv. Mater.* **2024**, *36* (30), No. 2401192.
- (6) Yang, S.; He, F.; Mai, K.; Zhou, Y.; Shang, C.; Liu, H.; Feng, Y.; Yu, G.; Li, J. Dynamic host-guest/ π - π multiple bonds of alginate-based super-amphiphiles reinforced the mechanical stability and foliar

deposition of jammed emulsion droplet. *Chem. Eng. J.* **2025**, *505*, No. 159377.

(7) Li, N.; Sun, C.; Jiang, J.; Wang, A.; Wang, C.; Shen, Y.; Huang, B.; An, C.; Cui, B.; Zhao, X.; et al. Advances in controlled-release pesticide formulations with improved efficacy and targetability. *J. Agric. Food Chem.* **2021**, *69* (43), 12579–12597.

(8) Zeng, Y.; Li, X.; Chen, F.; Ye, H.; Rong, K.; Ran, Z.; Liu, B.; Pan, Z.; Xie, X.; Tang, J.; et al. Biomineralization-inspired porous calcium carbonate microspheres as a controlled release system of herbicides/pesticides. *J. Environ. Chem. Eng.* **2024**, *12* (3), No. 112572.

(9) Zhang, X.; Wu, T.; Wang, L.; Liu, S.; Gao, Y.; Zhang, P.; He, C.; Gao, Y.; Zhang, C.; Zhang, L.; et al. Pesticide-fertilizer synergistic alginate-based hydrogel for enhanced pesticide retention and nutrient optimization. *Int. J. Biol. Macromol.* **2025**, *307*, No. 141877.

(10) Smagin, A.; Sadovnikova, N.; Smagina, M. Synthetic gel structures in soils for sustainable potato farming. *Sci. Rep.* **2019**, *9* (1), 18588.

(11) Liu, Y.; Wang, J.; Chen, H.; Cheng, D. Environmentally friendly hydrogel: A review of classification, preparation, and application in agriculture. *Sci. Total Environ.* **2022**, *846*, No. 157303.

(12) Wang, H.; Qu, G.; Liu, X.; He, M.; Yin, C.; Xu, R. Hydrogel materials in agriculture: A review. *J. Environ. Chem. Eng.* **2025**, *13* (3), No. 116385.

(13) Hu, H.; Li, P.; Tong, Y.; Li, J.; He, Y. Reutilization recovered nutrients from wastewater by synthesizing eco-friendly slow-release hydrogel fertilizer for sustainable vegetable production. *Chem. Eng. J.* **2024**, *499*, No. 155943.

(14) Wu, J.; Sheng, X.; Li, L.; Liang, J.; Li, Y.; Zhao, Z.; Cui, F. Rational design of a multifunctional hydrogel trap for water and fertilizer capture: a review. *J. Agric. Food Chem.* **2024**, *72* (31), 17176–17190.

(15) Li, Z.; Zhang, M. Progress in the preparation of stimulus-responsive cellulose hydrogels and their application in slow-release fertilizers. *Polymers.* **2023**, *15* (17), 3643.

(16) Rebelo, R.; Fonseca, A.; Coelho, J.; Serra, A. A sustainable synthesis of cellulose hydrogels for agriculture with repurpose of solvent as fertilizer. *Carbohydr. Polym.* **2025**, *368*, No. 124156.

(17) Sarhan, N.; Arafa, E.; Elgiddawy, N.; Elsayed, K.; Mohamed, F. Urea intercalated encapsulated microalgae composite hydrogels for slow-release fertilizers. *Sci. Rep.* **2024**, *14* (1), 15032.

(18) Cheng, D.; Liu, Y.; Yang, G.; Zhang, A. Water- and fertilizer-integrated hydrogel derived from the polymerization of acrylic acid and urea as a slow-release N fertilizer and water retention in agriculture. *J. Agric. Food Chem.* **2018**, *66* (23), 5762–5769.

(19) Xu, C.; Cao, L.; Bilal, M.; Cao, C.; Zhao, P.; Zhang, H.; Huang, Q. Multifunctional manganese-based carboxymethyl chitosan hydrogels for pH-triggered pesticide release and enhanced fungicidal activity. *Carbohydr. Polym.* **2021**, *262*, No. 117933.

(20) Yan, X.; Wang, T.; Li, H.; Zhang, L. N.; Xin, H.; Lu, G. Y. Flexible aggregation-induced emission-active hydrogel for on-site monitoring of pesticide degradation. *ACS Nano* **2022**, *16* (11), 18421–18429.

(21) Chaloner, T.; Gurr, S.; Bebbler, D. P. Plant pathogen infection risk tracks global crop yields under climate change. *Nat. Clim. Change.* **2021**, *11* (8), 710–715.

(22) Delgado-Baquerizo, M.; Guerra, C.; Cano-Díaz, C.; Egidi, E.; Wang, J.; Eisenhauer, N.; Singh, B.; Maestre, F. The proportion of soil-borne pathogens increases with warming at the global scale. *Nat. Clim. Change.* **2020**, *10* (6), 550–554.

(23) Lin, C.; Hur, J.; Zhang, M.; Zhang, Y.; Zhang, L.; Huang, Z.; Liu, X.; Li, C.; Huang, J.; Chu, F.; et al. Alleviating heat stress in cultivated plants with a radiative cooling and moisturizing film. *Energy Convers. Manage.* **2024**, *315*, No. 118786.

(24) Chen, Y.; Chen, Y. Thermo and pH-responsive methylcellulose and hydroxypropyl methylcellulose hydrogels containing K_2SO_4 for water retention and a controlled-release water-soluble fertilizer. *Sci. Total Environ.* **2019**, *655*, 958–967.

(25) Narayana, S.; Gowda, B.; Hani, U.; Ahmed, M.; Asiri, Z.; Paul, K. Smart poly(N-isopropylacrylamide)-based hydrogels: a tour D'hORIZON of biomedical applications. *Gels.* **2025**, *11* (3), 207.

(26) Wang, X.; Zheng, Y. Temperature-induced algae harvesting with grafted poly (N-isopropylacrylamide). *Bioresour. Technol.* **2025**, *438*, No. 133196.

(27) Gonzalez-Urias, A.; Licea-Claverie, A.; Sañudo-Barajas, J.; González-Ayón, M. NVCL-based hydrogels and composites for biomedical applications: progress in the last ten years. *Int. J. Mol. Sci.* **2022**, *23* (9), 4722.

(28) Zhuo, S.; Shu Hieng Tie, B.; Keane, G.; Geever, L. Strategies for developing shape-shifting behaviours and potential applications of poly (N-vinyl caprolactam) hydrogels. *Polymers* **2023**, *15* (6), 1511.

(29) Shang, H.; Yang, X.; Liu, H. Temperature-responsive hydrogel prepared from carboxymethyl cellulose-stabilized N-vinylcaprolactam with potential for fertilizer delivery. *Carbohydr. Polym.* **2023**, *313*, No. 120875.

(30) Lin, X.; Guo, L.; Shaghaleh, H.; Hamoud, Y.; Xu, X.; Liu, H. A TEMPO-oxidized cellulose nanofibers/MOFs hydrogel with temperature and pH responsiveness for slow-release. *Int. J. Biol. Macromol.* **2021**, *191*, 483–491.

(31) Yang, X.; Li, Z.; Liu, H.; Ma, L.; Huang, X.; Cai, Z.; Xu, X.; Shang, S.; Song, Z. Cellulose-based polymeric emulsifier stabilized poly(N-vinylcaprolactam) hydrogel with temperature and pH responsiveness. *Int. J. Biol. Macromol.* **2020**, *143*, 190–199.

(32) Wang, J.; Liu, H.; Xu, X.; Guo, L.; Wang, D.; Shang, S.; Song, Z. Temperature-responsive hydrogel prepared from N-vinyl- ϵ -caprolactam Pickering emulsions stabilized by cationic cellulose nanocrystals for preservative delivery in food packaging. *Int. J. Biol. Macromol.* **2025**, *309*, No. 142847.

(33) Mostafalu, R.; Heydari, A.; Banaei, A.; Ghorbani, F.; Arefi, M. The use of palladium nanoparticles supported on active carbon for synthesis of disproportionate rosin (DPR). *J. Nanostruct. Chem.* **2017**, *7* (1), 61–66.

(34) Pang, S.; Chen, H.; Jiang, Z.; Song, B.; Xie, D.; Zhai, Z.; Cui, Z.; Gu, Y.; Pei, X. Water-in-oil emulsion gels stabilized by a low-molecular weight organogelator derived from dehydroabietic acid. *Langmuir* **2022**, *38* (19), 6049–6056.

(35) Yan, T.; Song, B.; Du, D.; Cui, Z.; Pei, X. Rosin-based chiral wormlike micelles: rheological behavior and its application in preparing ultrasmall gold nanoparticles. *J. Colloid Interface Sci.* **2020**, *579*, 61–70.

(36) Yang, X.; Jiang, Z.; Liu, H.; Zhang, H.; Xu, X.; Shang, S.; Song, Z. Performance improvement of rosin-based room temperature vulcanized silicone rubber using nanofiller fumed silica. *Polym. Degrad. Stab.* **2021**, *183*, No. 109422.

(37) Lin, C.; Li, Y.; Tang, W.; Zhou, S.; Rao, X. Facile construction of bio-based supramolecular hydrogels from dehydroabietic acid with a tricyclic hydrophenanthrene skeleton and stabilized gel emulsions. *Molecules* **2021**, *26* (21), 6526.

(38) Gu, S.; Liu, M.; Xu, R.; Han, X.; Lou, Y.; Kong, Y.; Gao, Y.; Shang, S.; Song, Z.; Song, J.; et al. Ecofriendly controlled-release insecticide carrier: pH-/temperature-responsive rosin-derived hydrogels for avermectin delivery against *Mythimna separata* (Walker). *Langmuir* **2024**, *40* (21), 10992–11010.

(39) Hu, L.; Xie, Y.; Gao, S.; Shi, X.; Lai, C.; Zhang, D.; Lu, C.; Liu, Y.; Du, L.; Fang, X.; et al. Strain-induced orientation facilitates the fabrication of highly stretchable and tough xylan-based hydrogel for strain sensors. *Carbohydr. Polym.* **2023**, *312*, No. 120827.

(40) Dai, H.; Zhang, Y.; Ma, L.; Zhang, H.; Huang, H. Synthesis and response of pineapple peel carboxymethyl cellulose-g-poly (acrylic acid-co-acrylamide)/graphene oxide hydrogels. *Carbohydr. Polym.* **2019**, *215*, 366–376.

(41) Zhang, X.; Wu, T.; Wang, L.; Liu, S.; Gao, Y.; Zhang, P.; He, C.; Gao, Y.; Zhang, C.; Zhang, L.; et al. Pesticide-fertilizer synergistic alginate-based hydrogel for enhanced pesticide retention and nutrient optimization against apple Valsa canker. *Int. J. Biol. Macromol.* **2025**, *307*, No. 141877.

- (42) Zhang, L.; Chen, C.; Zhang, J.; Liu, B.; Teng, G.; Wang, J.; Zhang, X.; Cai, D.; Wu, Z. Alternating magnetic field-responsive nanoplatforms for controlled imidacloprid release and sustainable pest control. *ACS Sustain. Chem. Eng.* **2021**, *9* (31), 10491–10502.
- (43) Bera, B.; Khazal, R.; Schroën, K. Coalescence dynamics in oil-in-water emulsions at elevated temperatures. *Sci. Rep.* **2021**, *11* (1), 10990.
- (44) Zhang, J.; Wu, Q.; Li, M.; Song, K.; Sun, X.; Lee, S.; Lei, T. Thermoresponsive copolymer poly(N-vinylcaprolactam) grafted cellulose nanocrystals: synthesis, structure, and properties. *ACS Sustain. Chem. Eng.* **2017**, *5* (8), 7439–7447.
- (45) Lv, J.; Sun, B.; Jin, J.; Jiang, W. Mechanical and slow-released property of poly(acrylamide) hydrogel reinforced by diatomite. *Mater. Sci. Eng. C* **2019**, *99*, 315–321.
- (46) Sun, W.; Liu, X.; Hua, W.; Wang, S.; Wang, S.; Yu, J.; Wang, J.; Yong, Q.; Chu, F.; Lu, C. Self-strengthening and conductive cellulose composite hydrogel for high sensitivity strain sensor and flexible triboelectric nanogenerator. *Int. J. Biol. Macromol.* **2023**, *248*, No. 125900.
- (47) Loyer, F.; Combrisson, A.; Omer, K.; Moreno-Couranjou, M.; Choquet, P.; Boscher, N. Thermoresponsive water-soluble polymer layers and water-stable copolymer layers synthesized by atmospheric plasma initiated chemical vapor deposition. *ACS Appl. Mater. Interfaces.* **2019**, *11* (1), 1335–1343.
- (48) Dai, H.; Huang, Y.; Huang, H. Eco-friendly polyvinyl alcohol/carboxymethyl cellulose hydrogels reinforced with graphene oxide and bentonite for enhanced adsorption of methylene blue. *Carbohydr. Polym.* **2018**, *185*, 1–11.
- (49) Dai, H.; Huang, H. Enhanced swelling and responsive properties of pineapple peel carboxymethyl cellulose-g-poly(acrylic acid-co-acrylamide) superabsorbent hydrogel by the introduction of carclazte. *J. Agric. Food Chem.* **2017**, *65* (3), 565–574.
- (50) Wang, Q.; Zhao, Y.; Xu, H.; Yang, X.; Yang, Y. Thermosensitive phase transition kinetics of poly(N-isopropylacryl amide-co-acrylamide) microgel aqueous dispersions. *J. Appl. Polym. Sci.* **2009**, *113* (1), 321–326.
- (51) Cortez-Lemus, N.; Licea-Claverie, A. Poly(N-vinylcaprolactam), a comprehensive review on a thermoresponsive polymer becoming popular. *Prog. Polym. Sci.* **2016**, *53*, 1–51.
- (52) Guo, L.; Wang, Y.; Wang, M.; Shaghaleh, H.; Hamoud, Y.; Xu, X.; Liu, H. Synthesis of bio-based MIL-100(Fe)@CNF-SA composite hydrogel and its application in slow-release N-fertilizer. *J. Clean. Prod.* **2021**, *324*, No. 129274.
- (53) Liu, S.; Wu, Q.; Sun, X.; Yue, Y.; Tubana, B.; Yang, R.; Cheng, H. Novel alginate-cellulose nanofiber-poly(vinyl alcohol) hydrogels for carrying and delivering nitrogen, phosphorus and potassium chemicals. *Int. J. Biol. Macromol.* **2021**, *172*, 330–340.
- (54) Wu, T.; Zhao, K.; Zhang, C.; Zhong, T.; Li, Z.; Bao, Z.; Gao, Y.; Du, F. Promising delivery platform for smart pest control with high water-retaining capacity. *ACS Appl. Mater. Interfaces.* **2022**, *14* (49), 55062–55074.